

$$\begin{aligned}
C_G \rightarrow & \frac{1}{16\pi^2} \left(-\frac{1}{6} g_3^3 \text{LF}_{3,0}[m_3] + \frac{2}{5} g_3^3 \text{LF}_{5,-2}[m_3] - \frac{1}{18} \sum_p g_3^3 \text{LF}_{3,0}[m_{\tilde{d}}^p] + \right. \\
& \frac{1}{8} \sum_p g_3^3 \text{LF}_{4,-1}[m_{\tilde{d}}^p] - \frac{1}{15} \sum_p g_3^3 \text{LF}_{5,-2}[m_{\tilde{d}}^p] - \frac{1}{9} \sum_p g_3^3 \text{LF}_{3,0}[m_{\tilde{q}}^p] + \frac{1}{4} \sum_p g_3^3 \text{LF}_{4,-1}[m_{\tilde{q}}^p] - \\
& \left. \frac{2}{15} \sum_p g_3^3 \text{LF}_{5,-2}[m_{\tilde{q}}^p] - \frac{1}{18} \sum_p g_3^3 \text{LF}_{3,0}[m_{\tilde{u}}^p] + \frac{1}{8} \sum_p g_3^3 \text{LF}_{4,-1}[m_{\tilde{u}}^p] - \frac{1}{15} \sum_p g_3^3 \text{LF}_{5,-2}[m_{\tilde{u}}^p] \right)
\end{aligned}$$